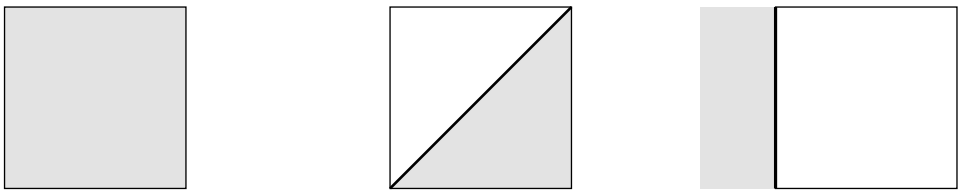
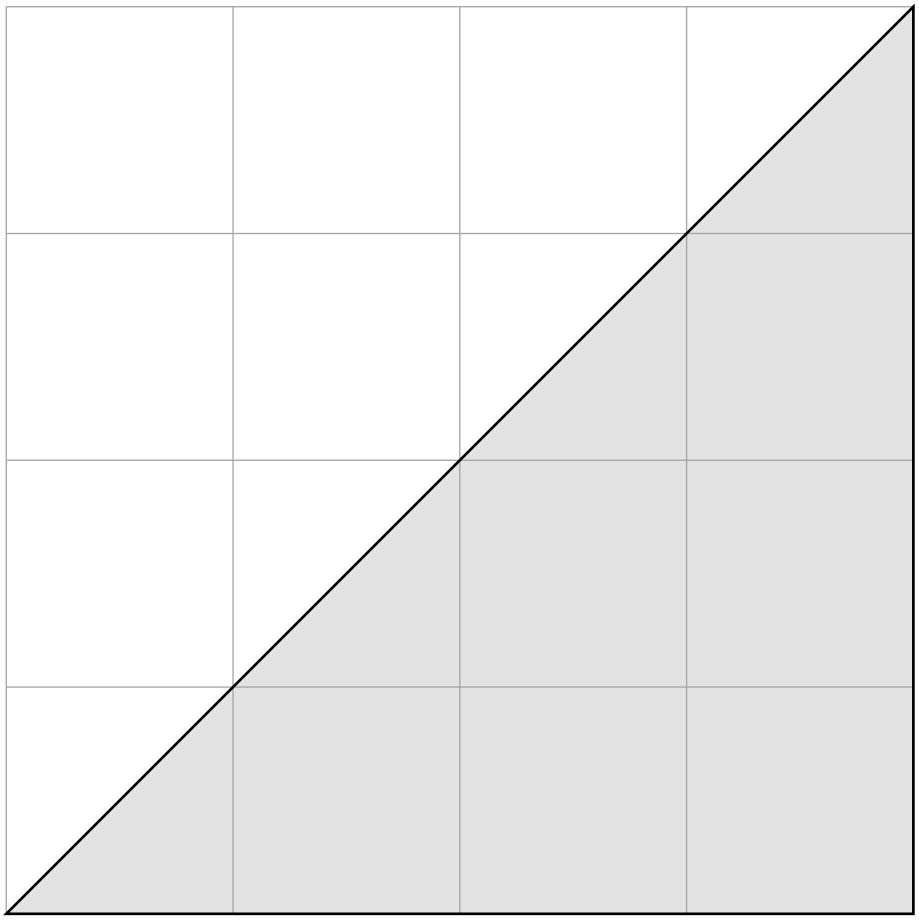
Whole squares are all gray. A cover uses every square with some gray in it. A touch at an edge or corner adds no square.

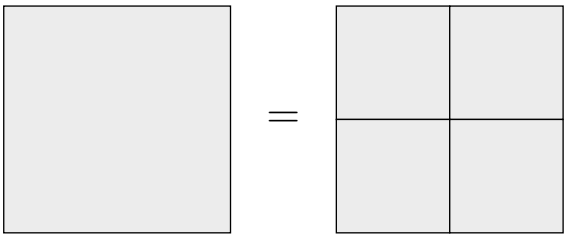


whole partial outside

Problem 1: Mark all the whole squares inside the triangle. Cover the triangle with as few grid-square tiles as you can.

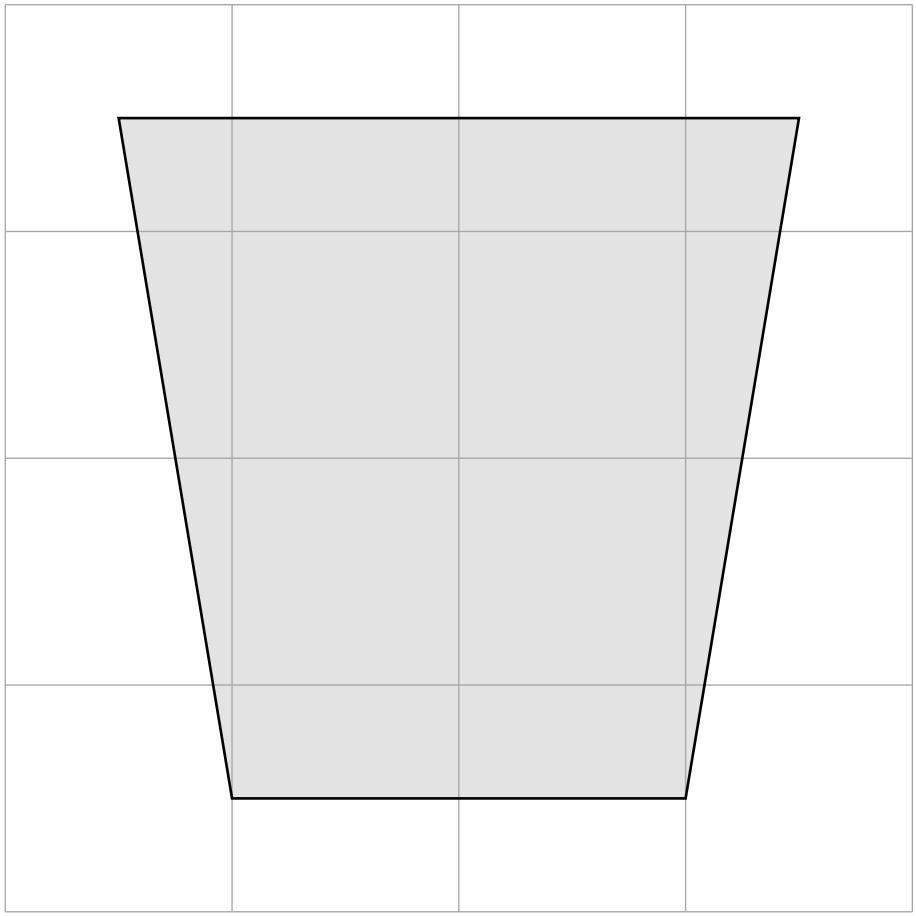


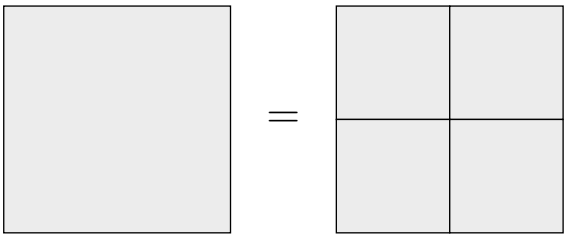
Inside: _____ big squares Cover: _____ big squares



1 big square
= 4 small squares

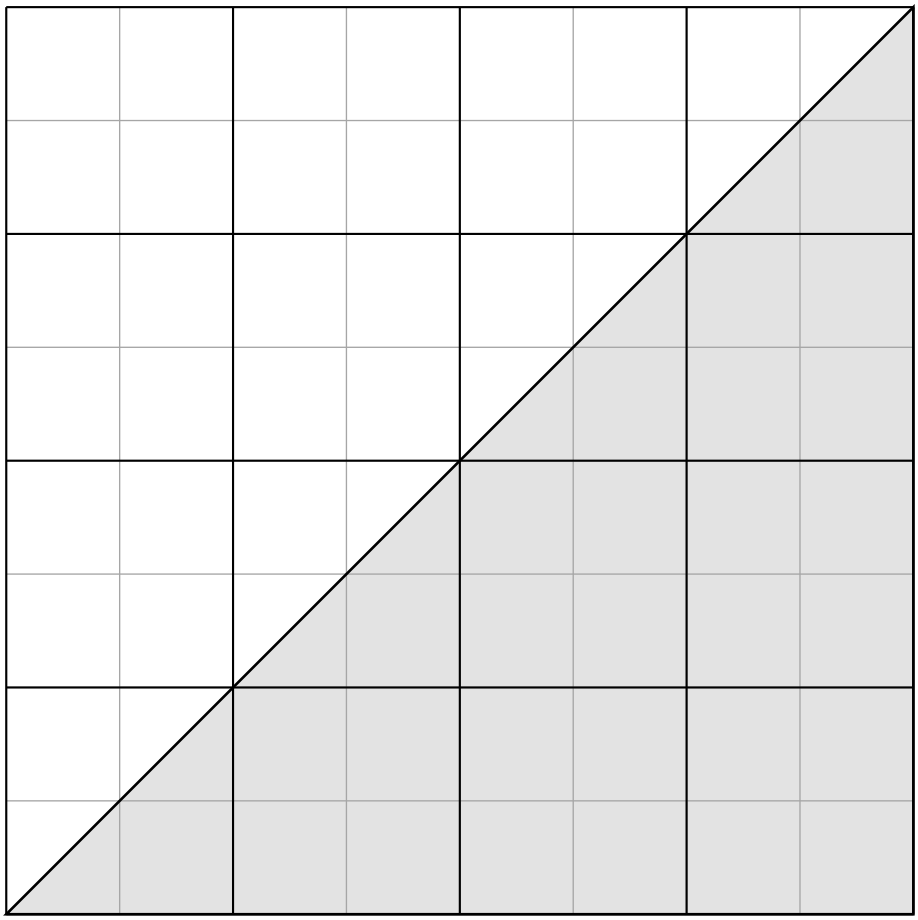
Problem 2: Cover this shape, then split two big tiles into four small tiles each. Choose the two big tiles that let you remove the most small tiles while keeping the gray shape covered.

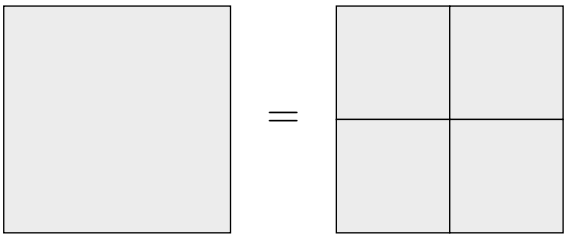




1 big square
= 4 small squares

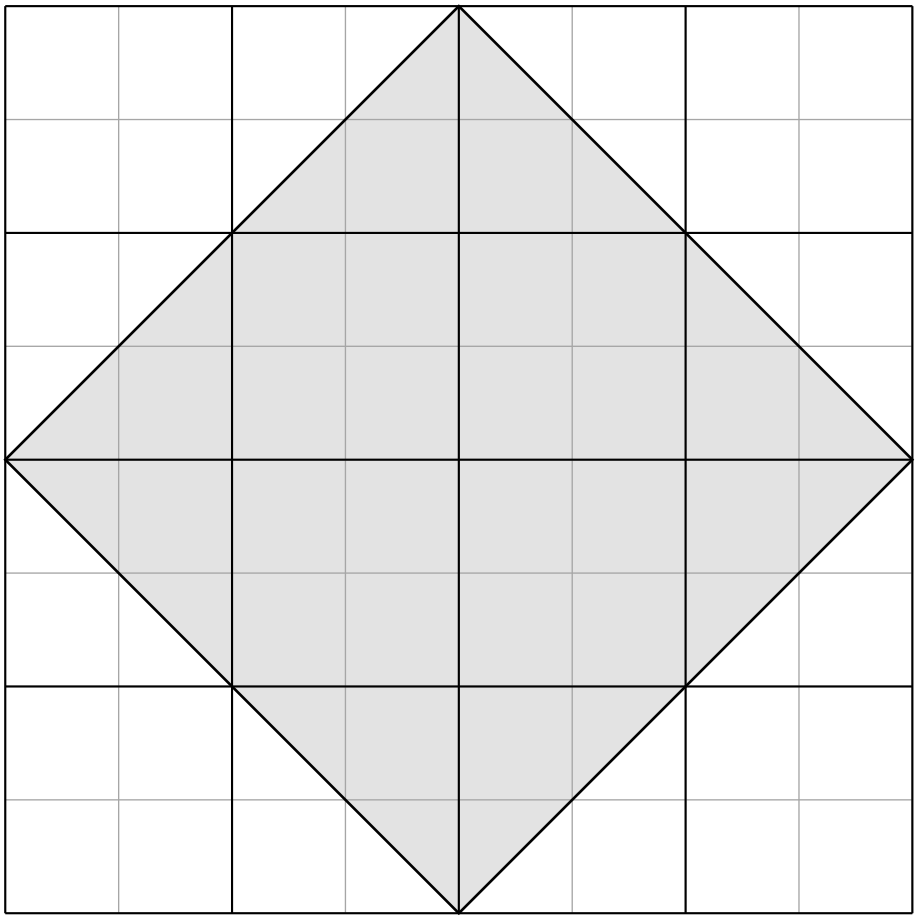
Problem 3: Do Problem 1 again with small squares.



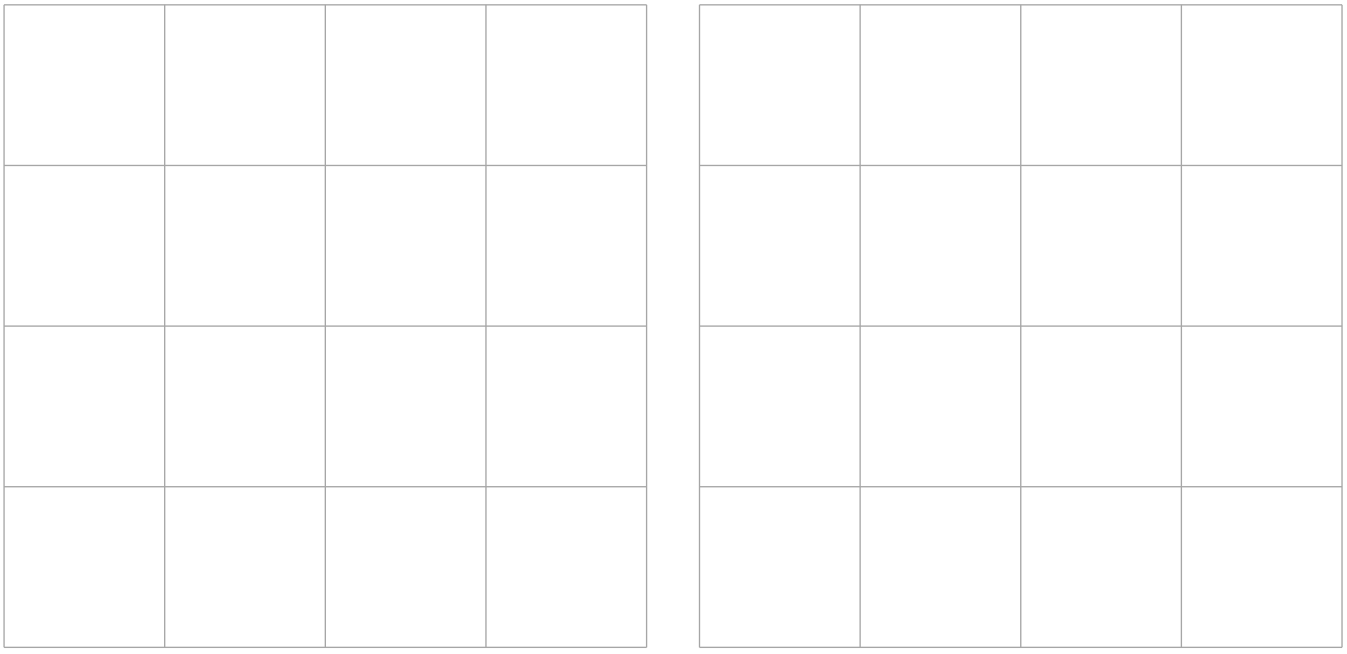


1 big square
= 4 small squares

Problem 4: Change the diamond so four outside small squares become partly gray. Keep every big square whole, partial, or outside as it was.



Problem 5: Draw two shapes, each with 3 whole and 3 partly gray squares. Make one with less than 4 big squares of gray and one with more than 5.



Problem 6: Keep the same whole and partial squares. Can you make your larger shape bigger again without filling another whole square?